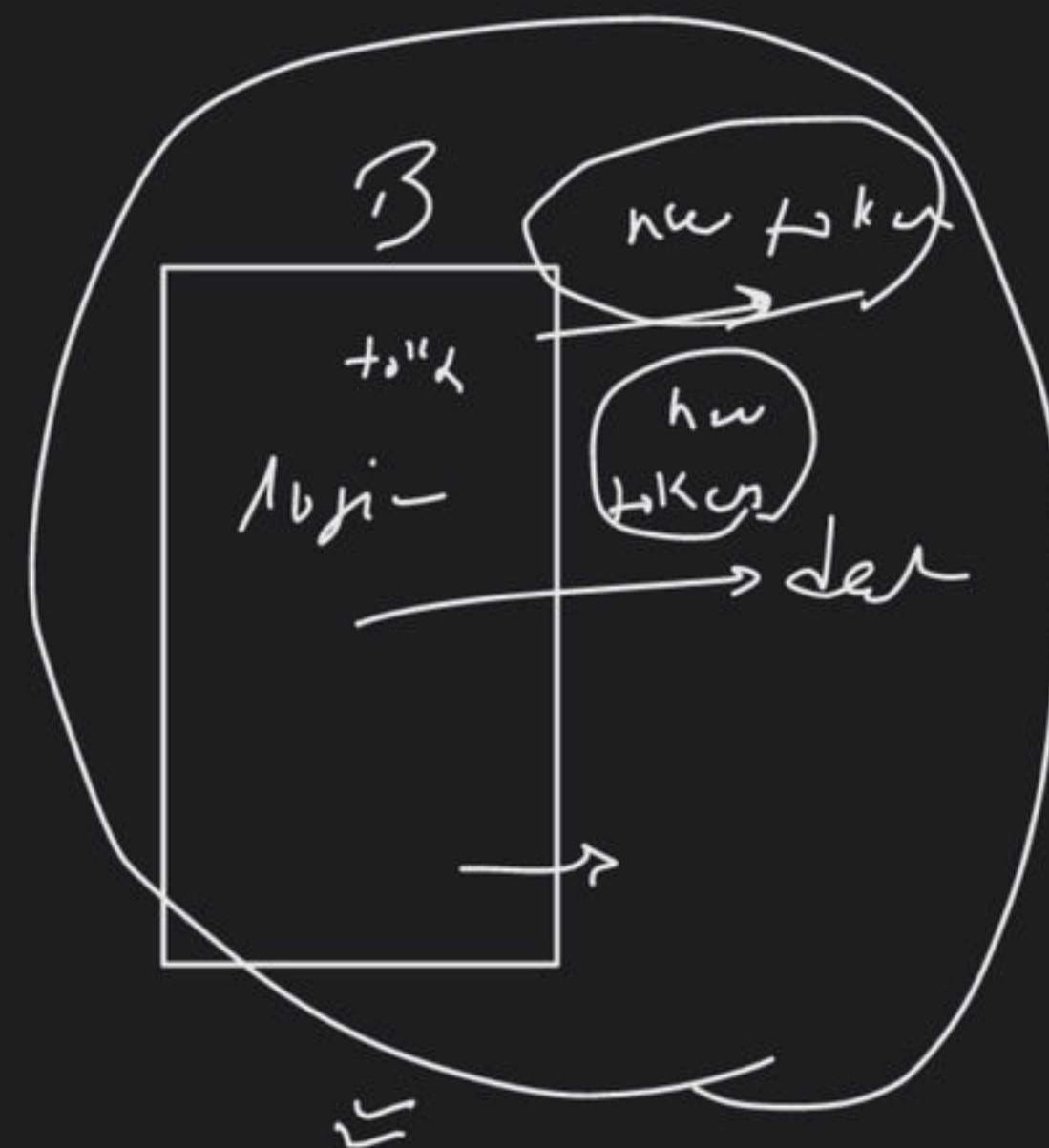
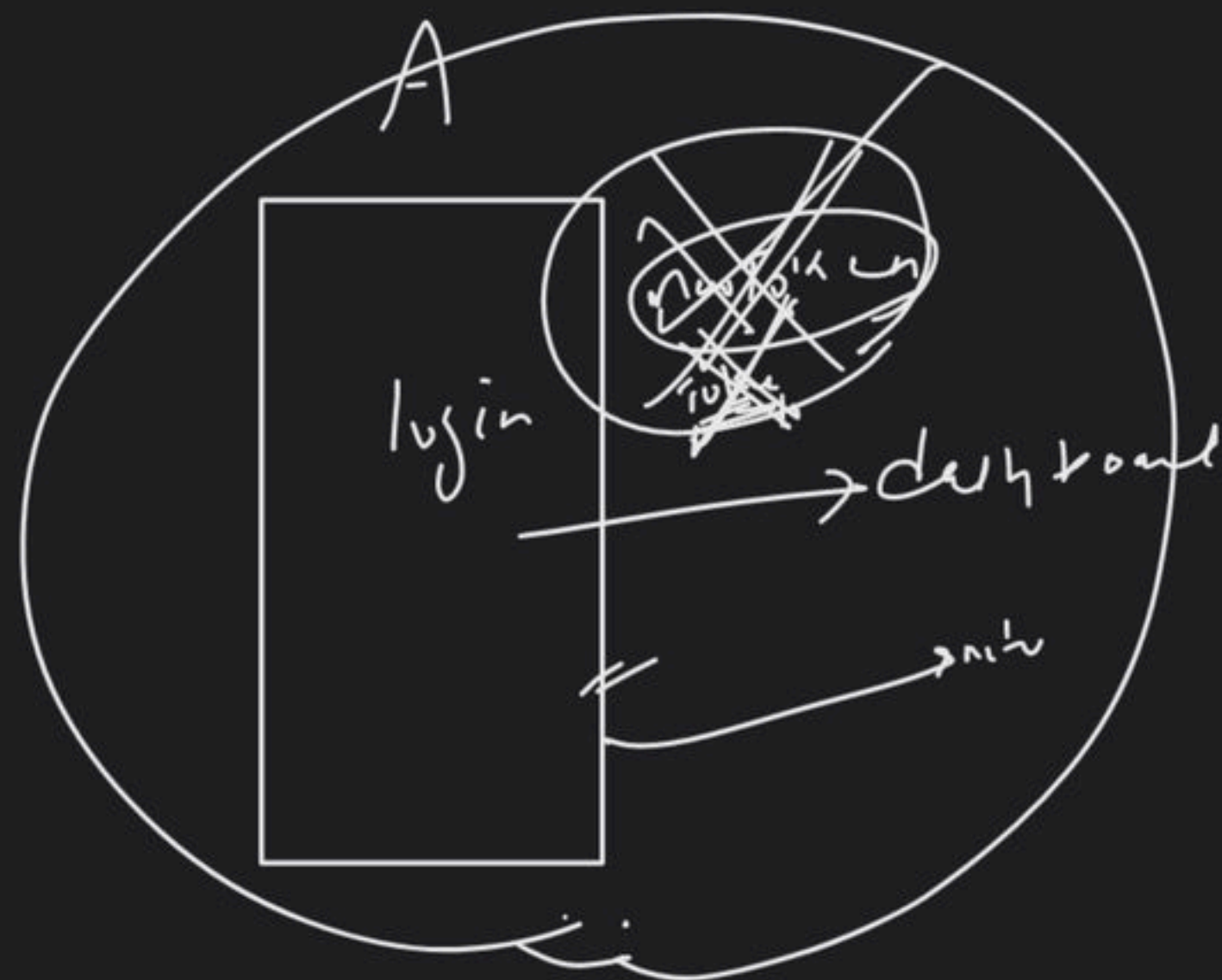


Heaps Class - 3

Special class

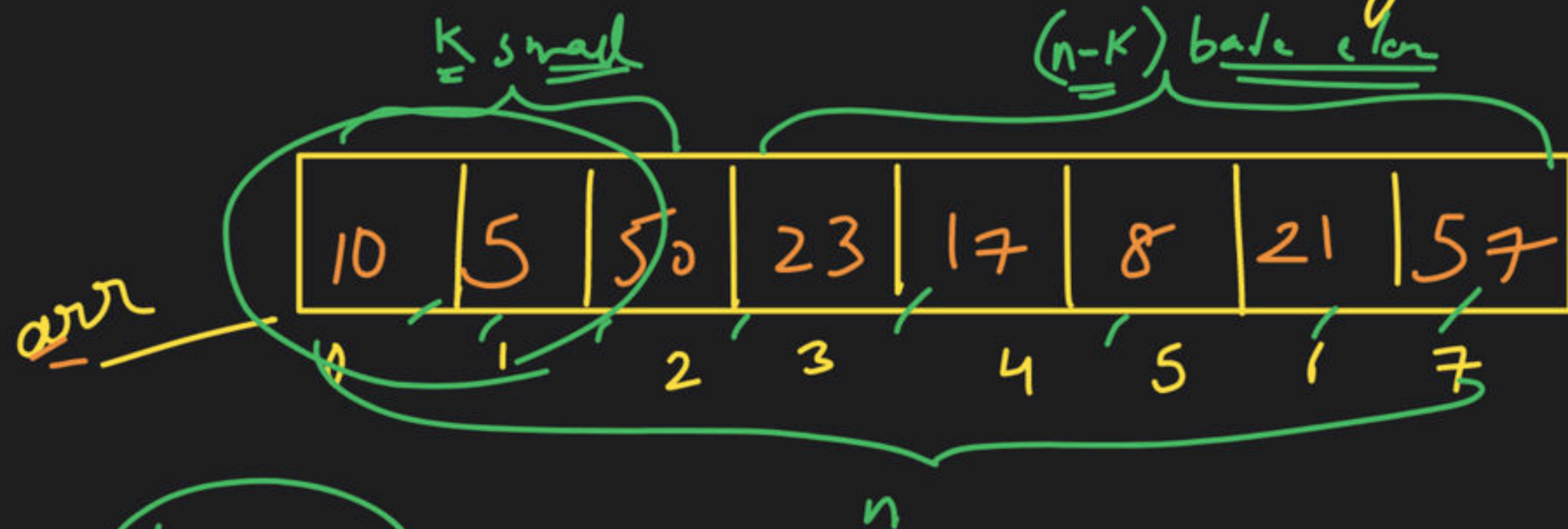


→ Map → Impl
heap'ly X
heapSort

→ STL - Priority-Queue → STL video
⇐
↓
Overview

→ K^{th} Smallest element in an array

$K=3$



Sort
 ↳ T.C → $n \log n$

Kth
 ↳ map
 ↳ ?

max-heap → K size → K small element → top

n size
 ↳ min heap

→ top → 1st small
 ↳ K pop → K^{th} smallest

min heap
 ↳ T.C → $O(n)$
 ↳ ~~$S.C. \rightarrow O(n)$~~

→ ans ✓

Max-heap

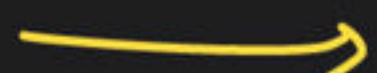
K-size

$K = 3$

K							
7	10	15	8	23	57	14	20
p	1	2	3	4	5	6	7



k^{th} largest elem



sort $\rightarrow$

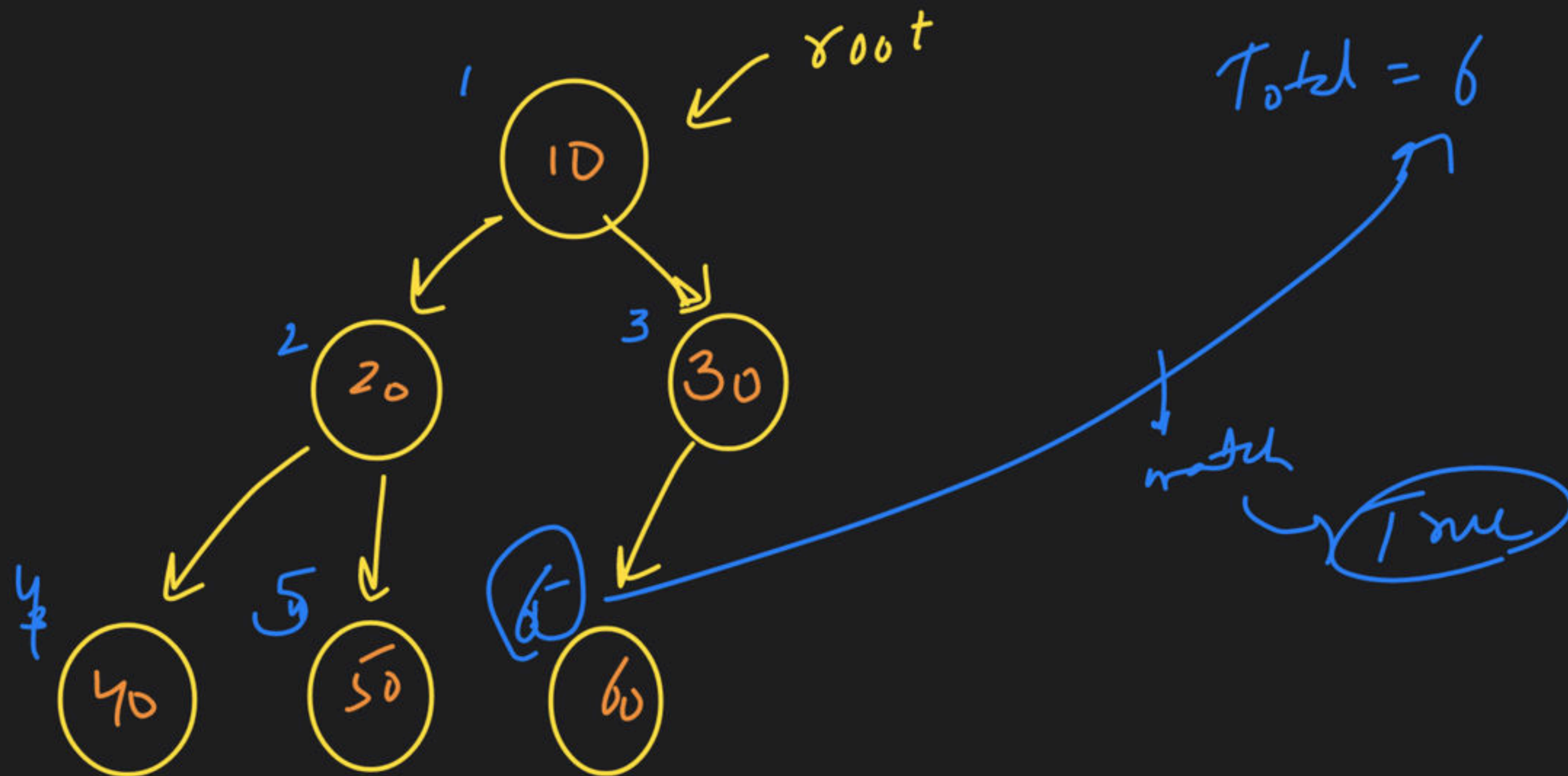


max-heap

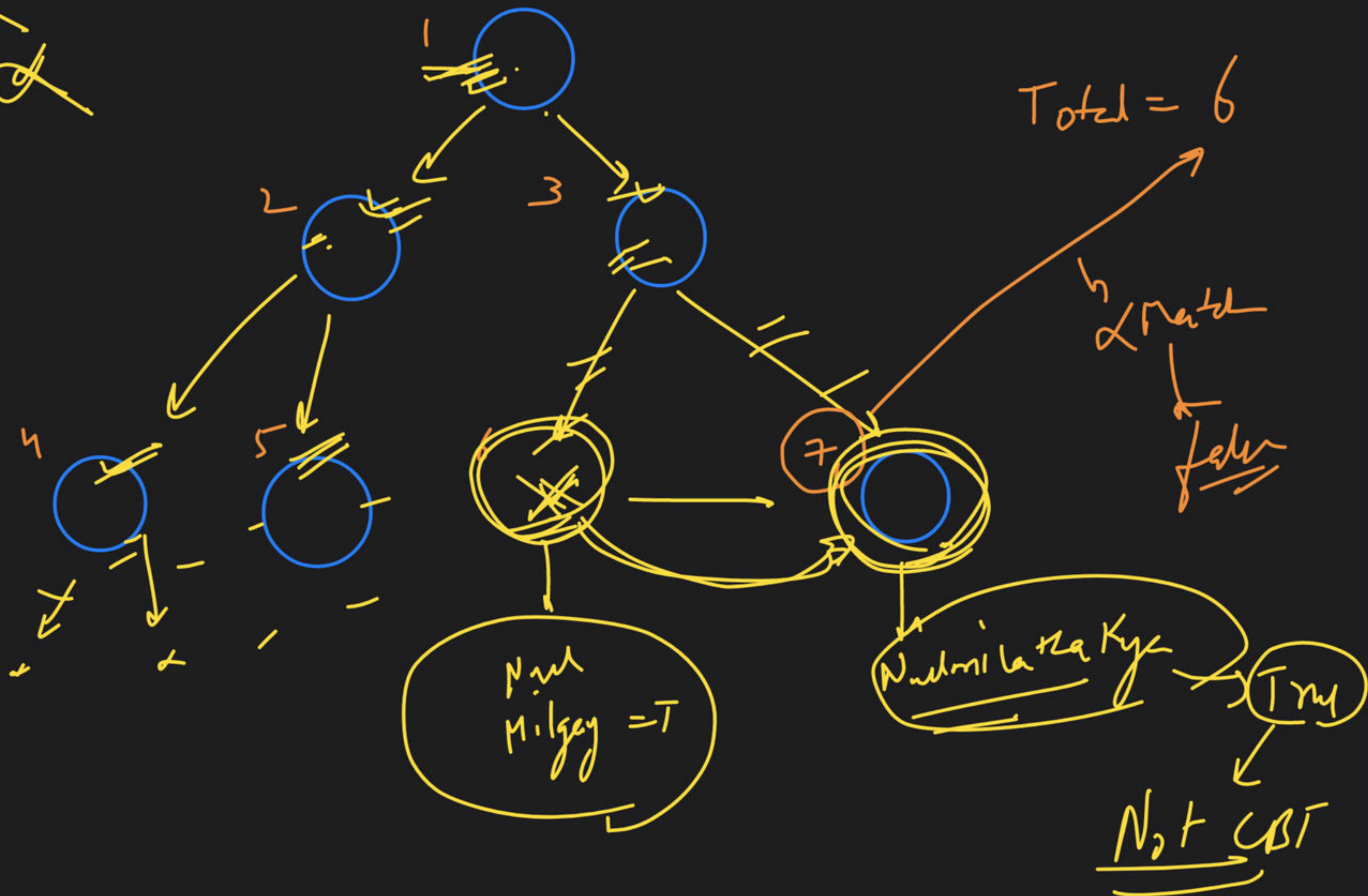


min-heap $\rightarrow$



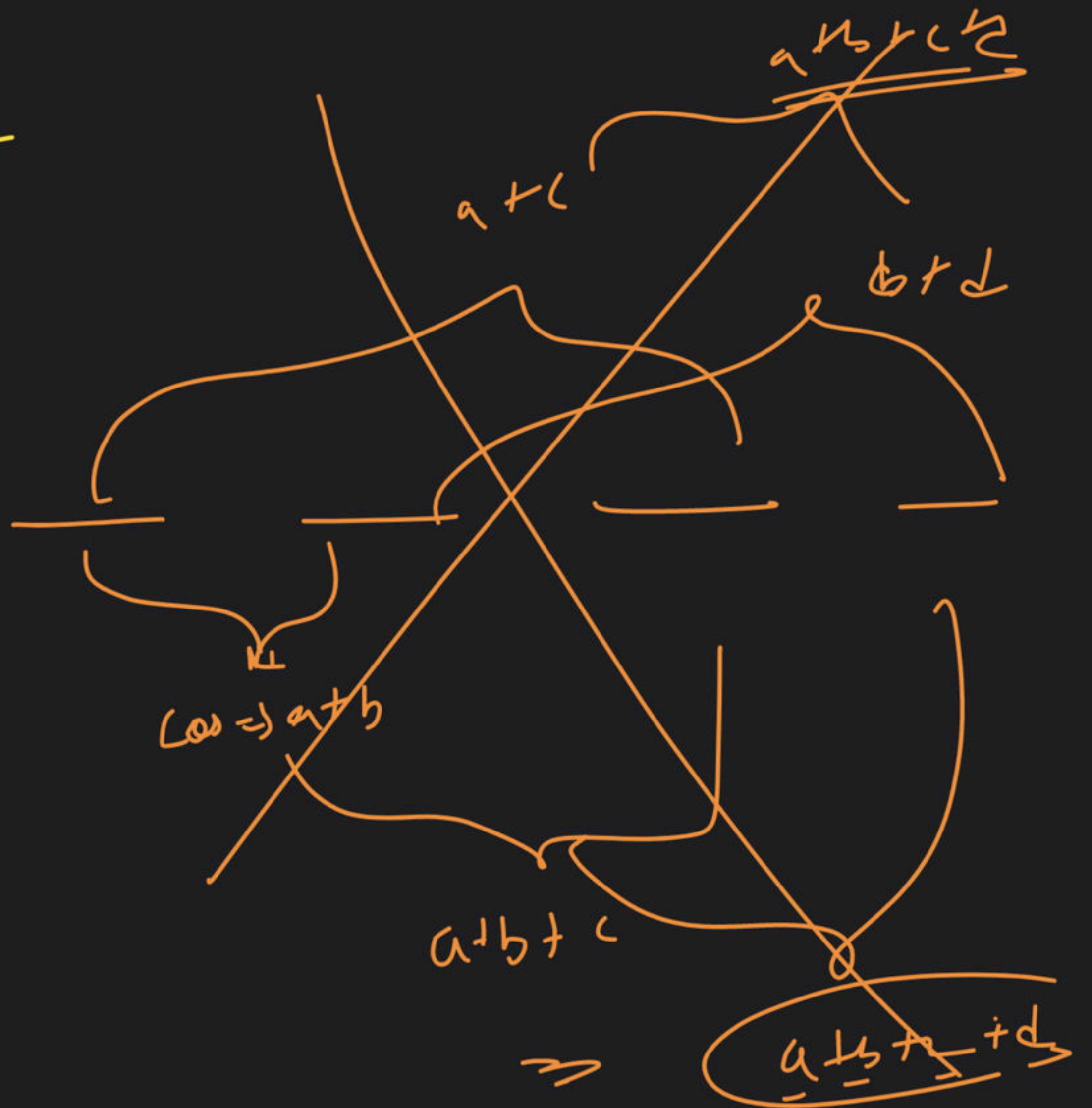


Count &
Numbering



→ Expression tree → $4/5 \rightarrow ?$

→ Connect ropes



$$a + b + c + d$$

$$a + b + \underbrace{c + d}_{\text{product}}$$

~~$$a + b + c + d$$~~

$$a + b$$

$$a + b + c$$

$$a + b + c + d$$

$$a + b$$

$$a + b + c + d$$

$$\begin{aligned} \text{Sum} &\downarrow \\ 0 &+ \\ a + b & \\ a + b + c & \\ a + b + c + d & \end{aligned}$$

$$3c + 3b + 2c + d$$

$$(2+9) \rightarrow \begin{matrix} 5 \\ 9 \\ 15 \end{matrix} \}, (29)$$

4	3	2	6
---	---	---	---

min-heap

